

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
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S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 902+200

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

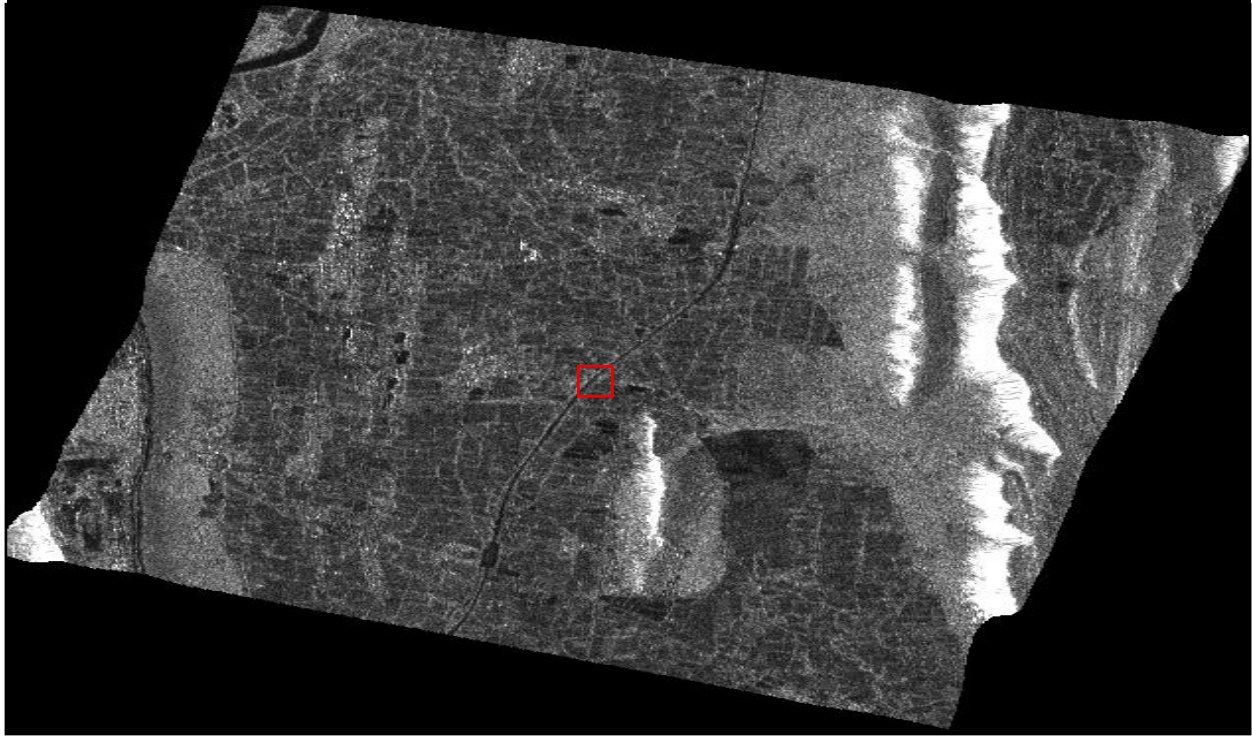
Flight Direction - DESCENDING

Polarization – VV+VH

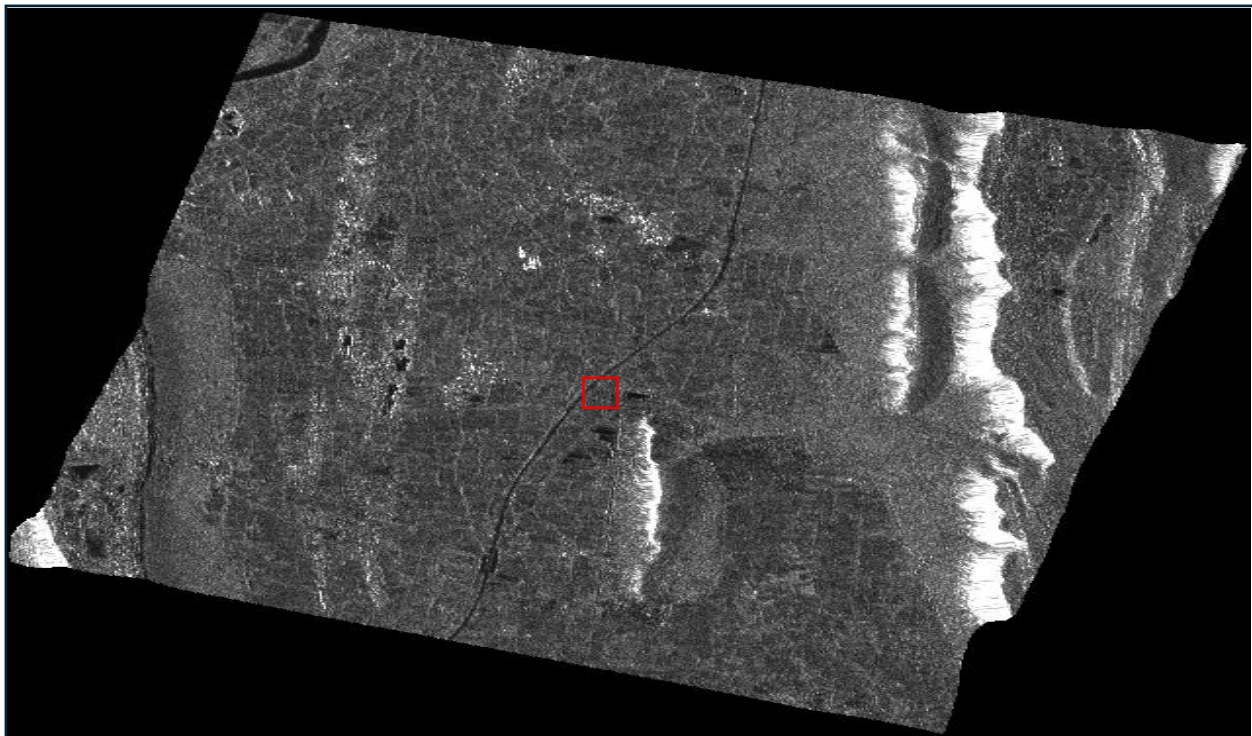
Absolute Orbit – 54706Frame – 519

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

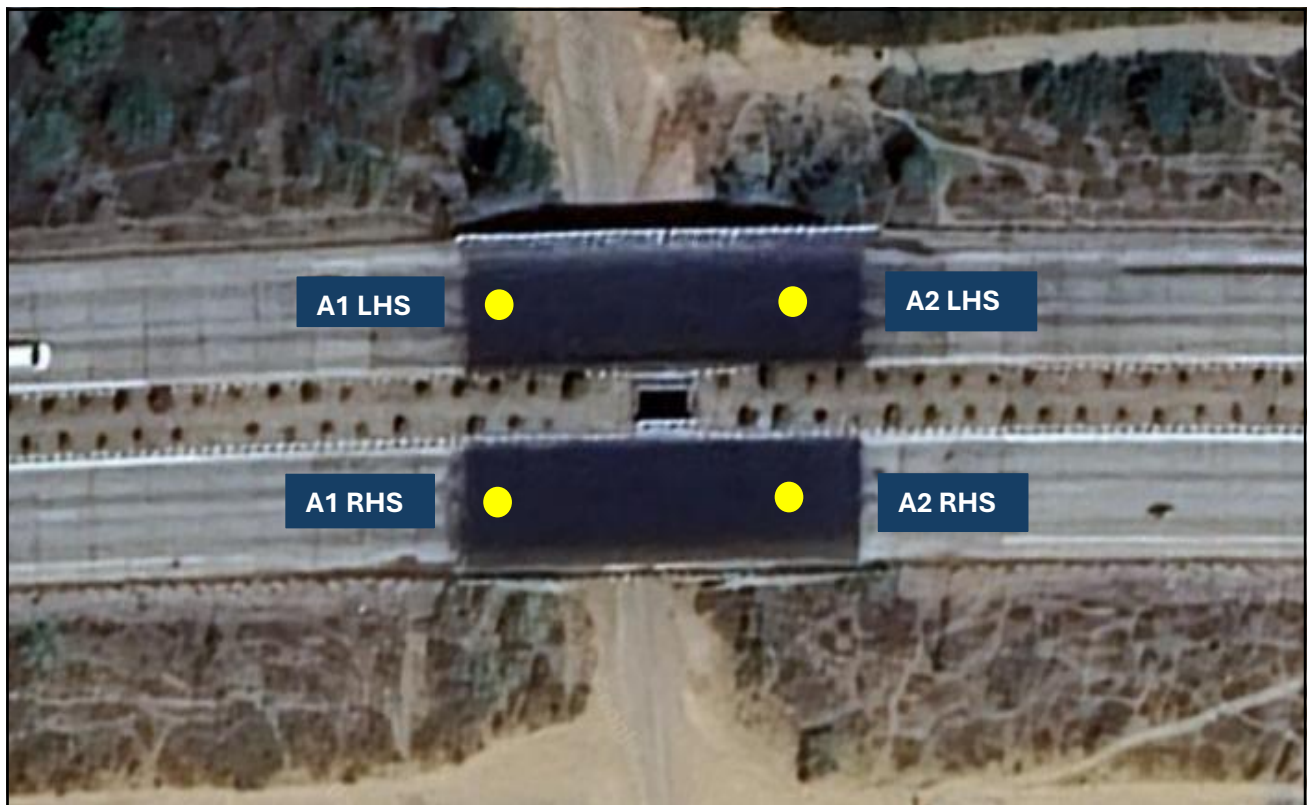


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

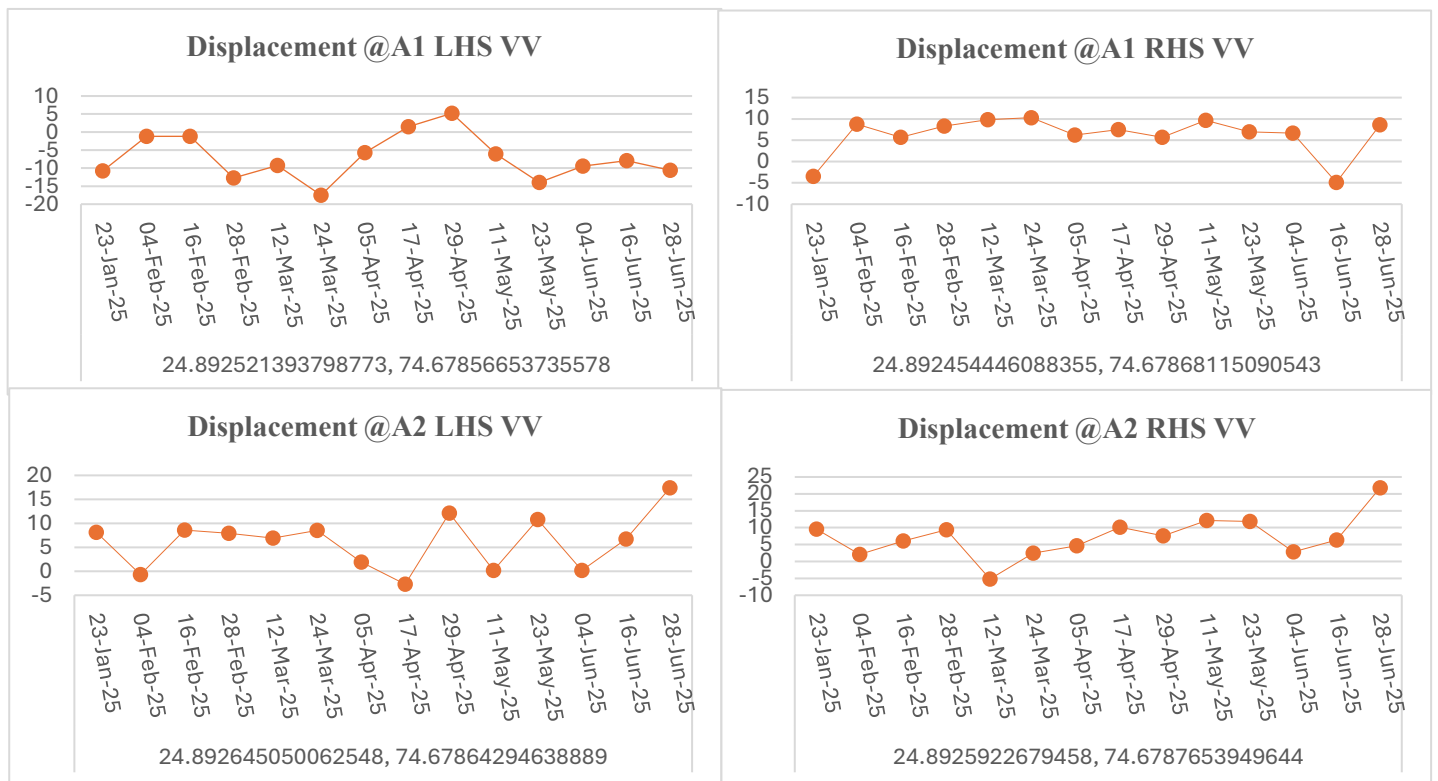
VUP – CH-906+250

Length – 31.53M

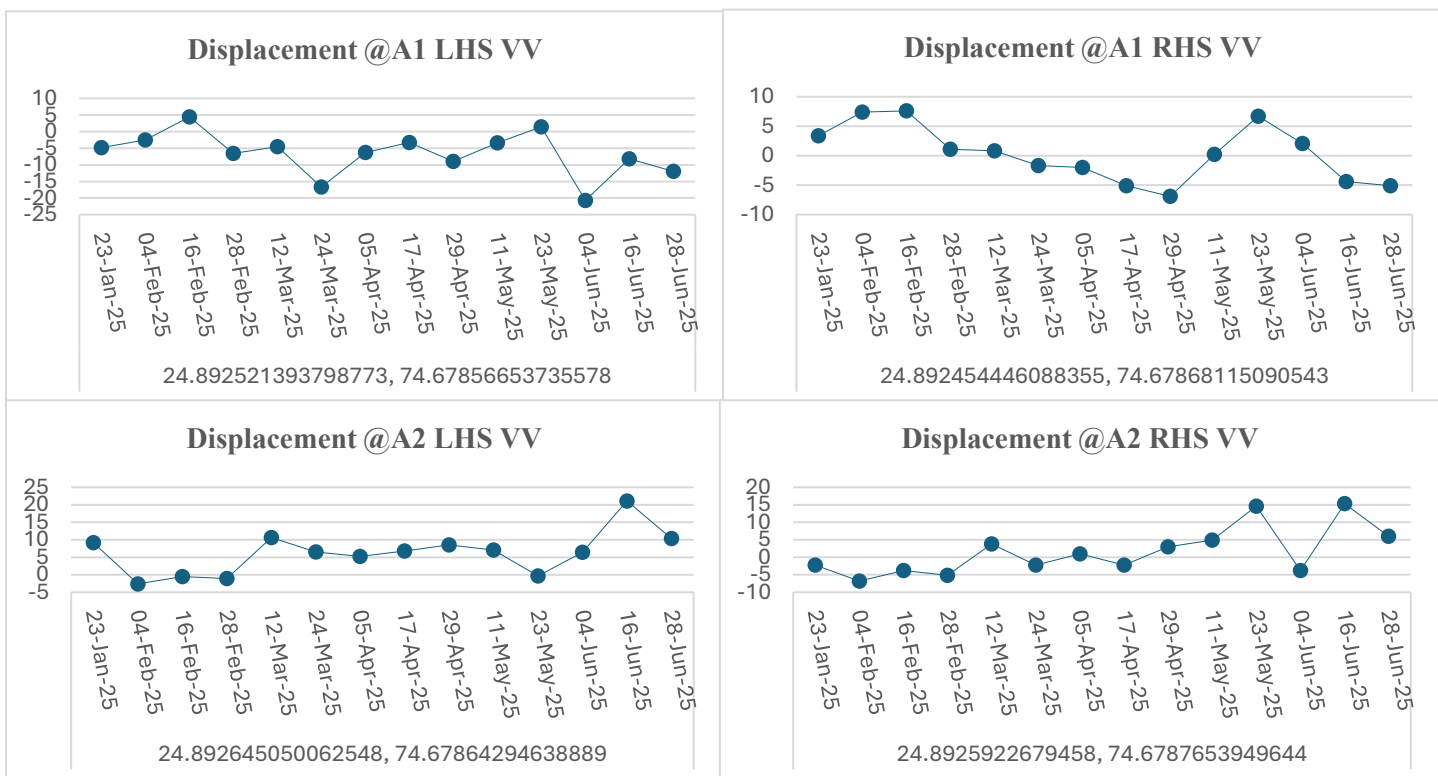


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



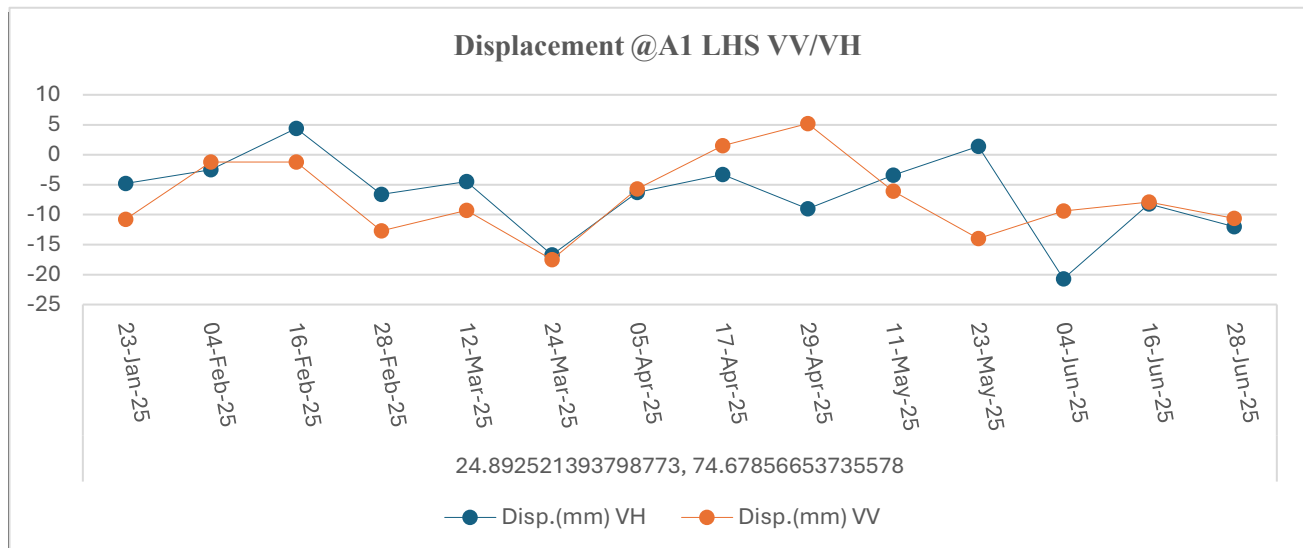
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.892521393798773, 74.67856653735578)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement exhibits moderate variability with an overall negative dominance, indicating prevailing vertical settlement at the A1 LHS location during the monitoring period.
- **Horizontal Displacement (VH)** Horizontal displacement shows significant variability, ranging roughly between +4 mm and –20 mm.

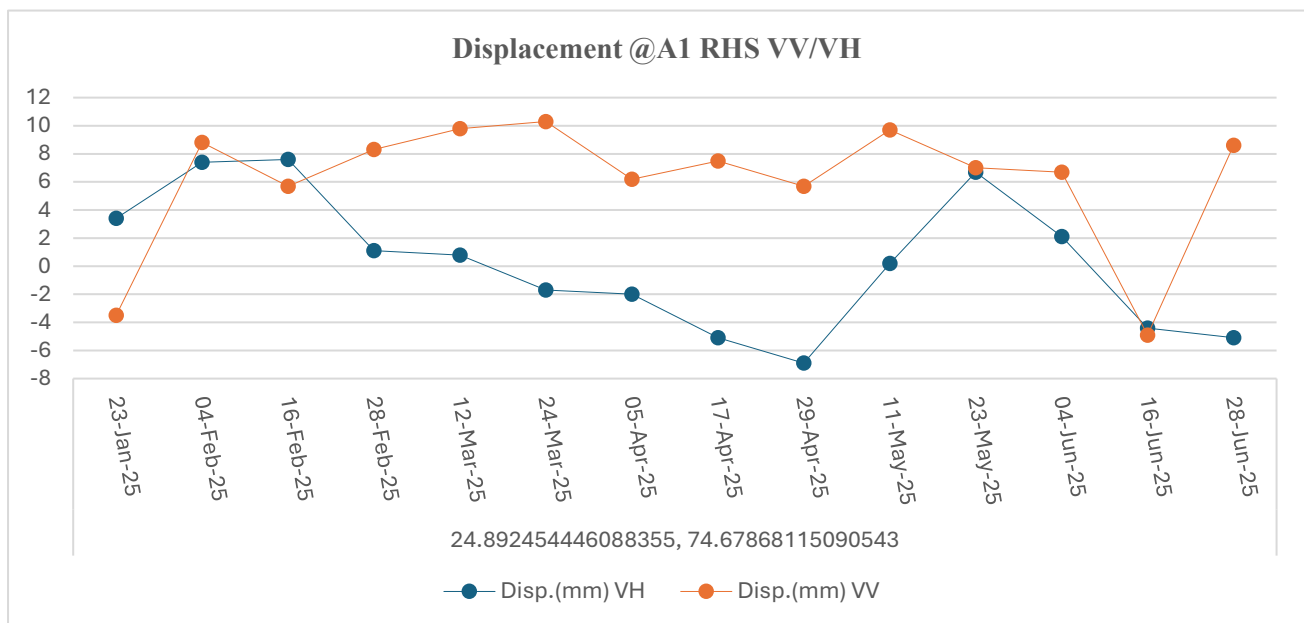
Critical Events & Anomalies:

- **February 2025:** VH improves slightly to –3 mm, while VV approaches –2 mm, suggesting a temporary reduction in settlement magnitude.
- **May 2025:** VH improves slightly to +1 mm, while VV drops to –14 mm, indicating vertical settlement dominance despite reduced lateral movement.
- **June 2025:** Both VH and VV remain negative (VH –12 mm, VV –11 mm), indicating no full recovery by the end of the monitoring period. Observation.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.892454446088355, 74.67868115090543)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows clear directional variability, ranging roughly between +7 mm and -7 mm.
- **Vertical displacement (VV)** exhibits moderate variability with a predominantly positive trend, indicating overall vertical uplift at the A1 RHS location during most of the monitoring period.

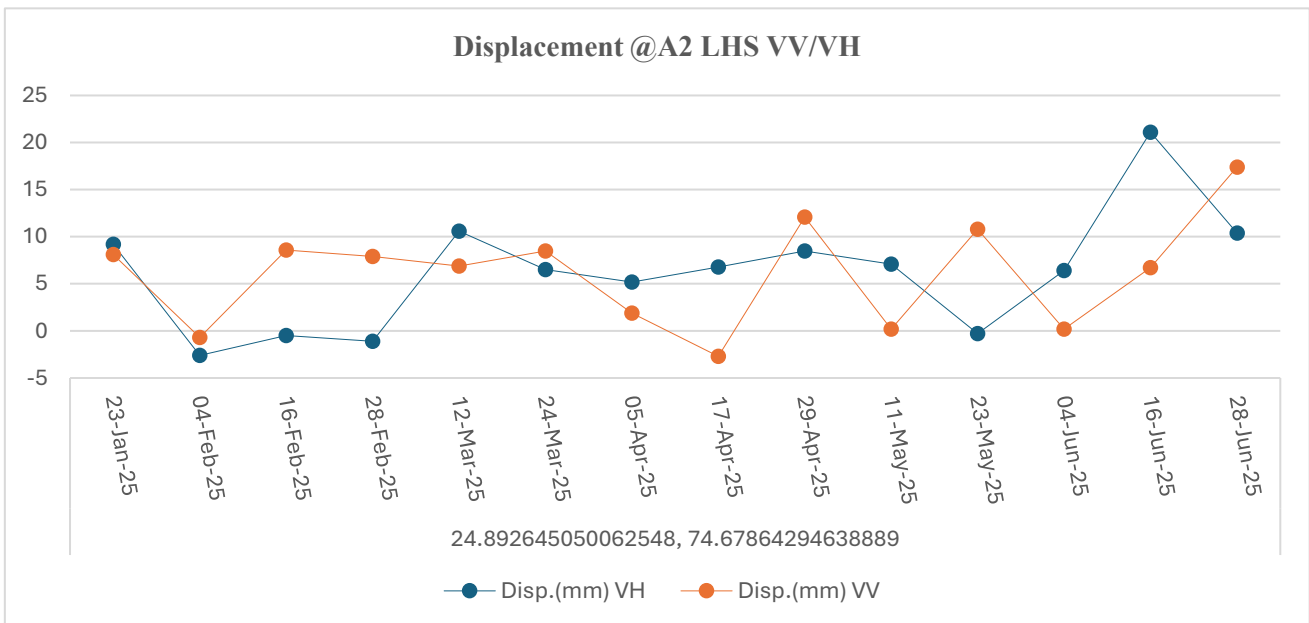
Critical Events & Anomalies:

- **February 2025:** Both VH and VV increase to +7 mm, representing a coincident uplift event affecting both vertical and horizontal components.
- **April 2025:** VH further decreases to -5 mm, while VV reduces to +6-7 mm, suggesting onset of lateral reversal without full vertical loss.
- **June 2025:** Both VH and VV drop sharply to -4 mm, marking the only strong combined settlement event during the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.892645050062548, 74.67864294638889)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows high variability with an overall increasing positive trend, ranging roughly between –3 mm and +21 mm.
- **Vertical Displacement (VV)** Vertical displacement exhibits moderate-to-high variability with alternating uplift and settlement phases over the monitoring period.

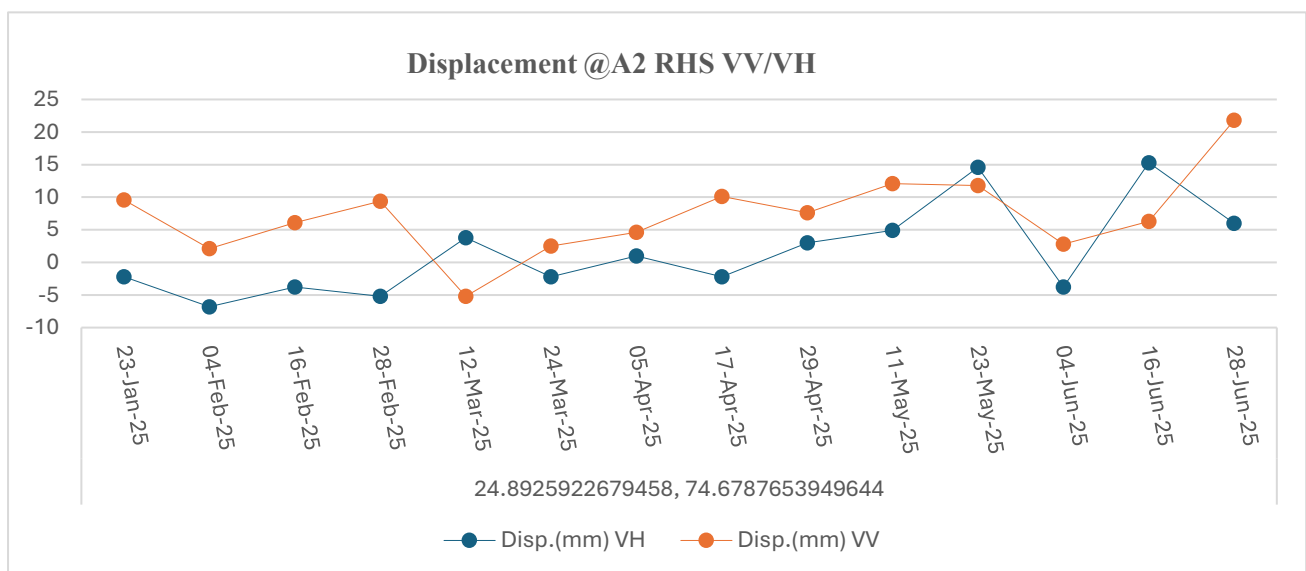
Critical Events & Anomalies:

- **February 2025:** VH drops to –2 mm, while VV approaches –1 mm, marking a short-lived coupled settlement.
- **March 2025:** VH rises sharply to +10 mm, while VV stabilises near +7 mm, representing a significant lateral uplift phase.
- **June 2025:** VH decreases slightly to +10 mm, while VV peaks near +18 mm, indicating strong vertical uplift toward the end of monitoring.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.8925922679458, 74.6787653949644)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement shows moderate-to-high variability with an overall increasing positive trend, indicating progressive vertical uplift at the A2 RHS.
- **Horizontal (VH) displacement** Horizontal displacement exhibits high variability with alternating lateral movement phases, ranging roughly between -7 mm and $+15$ mm.

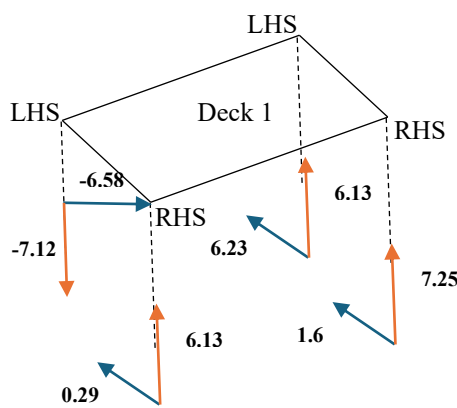
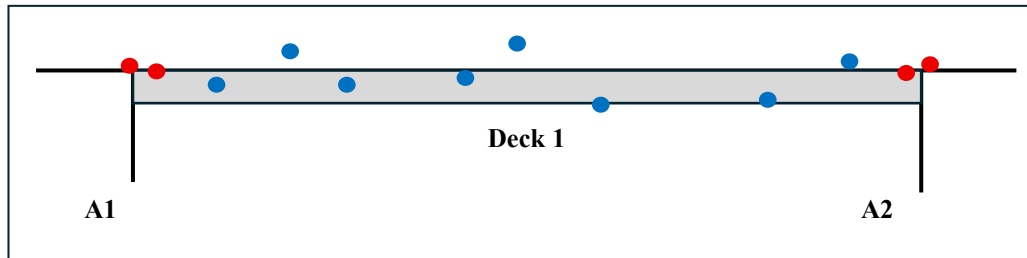
Critical Events & Anomalies:

- **February 2025:** VH drops to -7 mm, while VV reduces to $+3$ mm, marking a temporary lateral settlement event without loss of vertical uplift.
- **May 2025:** VH peaks near $+14$ mm, while VV stabilises around $+11$ mm, marking the largest lateral displacement event prior to June.
- **June 2025:** VH rebounds strongly to $+15$ mm, while VV increases sharply to $+22$ mm, representing the maximum uplift event of the entire monitoring period.

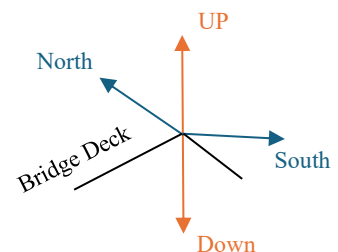
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits non-uniform vertical displacement, indicating differential vertical response between the LHS and RHS regions. The maximum downward settlement is observed at the LHS lower region, reaching approximately -7.12 mm, identifying localized settlement at the left support. The presence of uplift on the RHS and settlement on the LHS confirms vertical rotation of the deck slab, rather than uniform vertical deflection.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates clear directional variability, reflecting torsional deformation behaviour. The opposing directions and uneven magnitudes confirm torsional rotation rather than rigid-body translation.

CONCLUSION:

- The bridge deck slab demonstrates a torsion-influenced deformation pattern, characterized by asymmetric vertical uplift and settlement combined with differential horizontal displacement. Overall, the observed behaviour is moderate but structurally relevant, warranting continued monitoring and correlation with loading and boundary conditions.